

# ERNEST

Percussion oscillator with six ways to bend its pitch

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Ernest is a single percussion voice built around an oscillator whose pitch can be bent six different ways. Hit it with a trigger and you get kicks, toms, snares, zaps and metallic pings. Leave the trigger unpatched and it plays continuously, which is handy for tuning.

Beyond the classic drum-machine voice it has a few modular extras: a pitch input, CV over the modulation, a modulation output and a free-running mode.

## AT A GLANCE

- Sine or triangle oscillator, 20 Hz to 12 kHz, with a 1 V/oct input
- Six pitch modulation shapes, from a falling saw to a pitch envelope
- Modulation from 0.1 Hz to 5 kHz: slow wobbles turn into new tone colours
- One-knob decay envelope, bass boost with drive, ring modulation
- 12 HP

## CONTROLS

|                |                                                                                                                                                                                                                                                                                                                                              |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Screen         | The modulation shape and the base pitch.                                                                                                                                                                                                                                                                                                     |
| PITCH          | Base pitch, from 20 Hz to 12 kHz. It starts at 55 Hz.                                                                                                                                                                                                                                                                                        |
| WAVE           | Lit: triangle, slightly brighter. Off: sine.                                                                                                                                                                                                                                                                                                 |
| RETRIG         | Lit (default): every hit restarts the modulation, so every hit sounds the same. Off: the modulation keeps running and each hit lands somewhere else in its cycle.                                                                                                                                                                            |
| Shape selector | The six icons around this knob: <b>saw down</b> (the pitch falls, again and again), <b>square</b> (two pitches alternate), <b>triangle</b> (rises and falls), <b>random</b> (a new random step each cycle), <b>noise</b> (bursts of noise in the pitch, for snares) and <b>envelope</b> (one sweep down after each hit, for kicks and toms). |
| DEPTH          | How far the pitch moves, from -100 % to +100 %. Centre: no modulation. Negative values flip the direction. Gentle near the centre, up to $\pm 8$ octaves at the ends.                                                                                                                                                                        |
| RATE           | Modulation speed, from 0.1 Hz to 5 kHz. With the envelope shape, how fast the sweep falls back. Above about 20 Hz the modulation is heard as a new tone colour rather than movement.                                                                                                                                                         |
| DECAY          | Length of the volume envelope, from 3 ms to 3 s.                                                                                                                                                                                                                                                                                             |
| LEVEL          | Output level.                                                                                                                                                                                                                                                                                                                                |
| BASS           | Boosts the low end, then drives the sound into saturation as you turn it up.                                                                                                                                                                                                                                                                 |
| RING           | Lit: the oscillator is multiplied by the signal at RING IN. Nothing comes out unless both are sounding.                                                                                                                                                                                                                                      |

## INPUTS

|         |                                                                                                                                                                       |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TRIG    | Plays a hit: restarts the oscillator, both envelopes and, with RETRIG lit, the modulation. The small light flashes on each hit. Unpatched, Ernest plays continuously. |
| V/OCT   | Added to PITCH, 1 volt per octave.                                                                                                                                    |
| DEPTH   | Added to the DEPTH knob; $\pm 5$ V covers the whole range.                                                                                                            |
| RATE    | Added to the RATE knob; 1 V moves it a tenth of its travel.                                                                                                           |
| RING IN | The signal used by ring modulation ( $\pm 5$ V).                                                                                                                      |

## OUTPUTS

|     |                                                                                                                  |
|-----|------------------------------------------------------------------------------------------------------------------|
| MOD | The pitch modulation itself, after DEPTH: 5 V for 8 octaves. Use it to move other modules in step with each hit. |
| OUT | Audio, $\pm 5$ V at full LEVEL.                                                                                  |

## HOW IT WORKS

### Which way the pitch moves

Saw down, square, triangle and envelope start from PITCH and move up from it (or down, with negative DEPTH), then come back. Random and noise move both ways around PITCH.

The envelope jumps to full on every hit and falls back at a speed set by RATE. Noise is strongest at the start of each modulation cycle and fades towards its end, which gives snares their crack.

## PATCH IDEAS

### 1. Analog kick

Ernest's starting settings: envelope shape, DEPTH 50 %, RATE at the centre, PITCH 55 Hz, DECAY around two o'clock. Patch a clock into TRIG.

### 2. Snare

Noise shape, WAVE lit, PITCH around 200 Hz, DEPTH 40 %, RATE far right, DECAY short, a little BASS.

### 3. Tuned toms

Envelope shape, DEPTH 30 %, a short DECAY, and a pitch sequence into V/OCT.

### 4. Metallic ping

Two Ernests triggered together. Patch the first OUT into the second RING IN and light RING on the second. Raise RATE on either for inharmonic overtones.

### 5. Zap

Envelope shape, DEPTH at 100 %, RATE low for a slow sweep, DECAY short.

## SPECIFICATIONS

|            |                                                                             |
|------------|-----------------------------------------------------------------------------|
| Width      | 12 HP                                                                       |
| Pitch      | 20 Hz to 12 kHz, plus 1 V/oct input                                         |
| Modulation | 0.1 Hz to 5 kHz, up to $\pm 8$ octaves                                      |
| Decay      | 3 ms to 3 s                                                                 |
| Levels     | Audio $\pm 5$ V. MOD output 5 V per 8 octaves. Trigger threshold about 1 V. |